

Profile

Compiler engineer with hands-on experience across the complete LLVM/Clang pipeline, with most of my work focused on the middle end. I combine broad knowledge of processor architectures with experience across the front end, analyses and transformations, code generation, custom backends, and optimizations tailored to specific hardware features. I am particularly interested in alias and memory-dependence analysis and transformations that unlock performance by restructuring code. These interests also shaped my PhD research.

Experience

GlobalFoundries

Remote

MEMBER OF TECHNICAL STAFF, SOFTWARE ENGINEERING

Jun. 2026 - present

- Continuing development of ASIP Designer's extensively customized LLVM/Clang-based C/C++ compiler after the ASIP business transferred from Synopsys to GlobalFoundries in June 2026.

Synopsys

Remote

STAFF SOFTWARE ENGINEER

Jan. 2023 - Jun. 2026

- Developed ASIP Designer's extensively customized LLVM/Clang-based compiler, a mature product codebase with more than a decade of product-specific development on top of upstream LLVM. ASIP Designer generates a C/C++ SDK and synthesizable RTL for an application-specific processor from its architecture description.
- Worked across the full compiler pipeline, including the Clang front end, LLVM middle end, and a customized backend that emits a proprietary format for downstream ASIP tools. Most development focused on middle-end analyses, transformations, and optimization-pipeline support.
- Also contributed to separate proprietary compiler infrastructure and backend tooling built around compilation models distinct from LLVM/GCC.
- Developed product-specific code-restructuring optimizations and identified further opportunities by profiling workloads, inspecting generated assembly, and relating bottlenecks to ASIP hardware requirements.
- Unlike a conventional fixed-target compiler, this compiler dynamically supports processor targets derived from architecture descriptions, ranging from RISC-V-based processors to complex, highly specialized DSP architectures.
- Contributed code and RFC work to ongoing LLVM upstream support for C **restrict**-qualified local pointer variables, extending the existing function-parameter support; co-led a roundtable on the proposal at an LLVM conference.

Delft University of Technology

Delft, the Netherlands

PHD CANDIDATE

Jan. 2019 - Dec. 2022

- Researched programmer-friendly intermittent computing, with work spanning compiler support, architecture, and user-facing systems.
- Published at PLDI, ASPLOS, and UbiComp/IMWUT.

Micro Elektronische Producten (MEP)

Landsmeer, the Netherlands

JUNIOR SOFTWARE DEVELOPER

Feb. 2014 - Sep. 2015

- Developed software for an in-house Texas Instruments DSP platform running an RTOS for Voice over IP applications, including SNMP support and a zero-copy messaging scheme between tasks.

Education

Delft University of Technology

Delft, the Netherlands

PHD · DISSERTATION: ACCELERATING PROGRAMMER-FRIENDLY INTERMITTENT COMPUTING · DEFENDED 29 JUN. 2023

Jan. 2019 - Jun. 2023

Delft University of Technology

Delft, the Netherlands

MSC EMBEDDED SYSTEMS (+ ONE-YEAR PRE-MASTER)

Sep. 2015 - Feb. 2019

Amsterdam University of Applied Sciences

Amsterdam, the Netherlands

BACHELOR ELECTRICAL ENGINEERING (HBO)

Sep. 2011 - Jul. 2015

Publications

Data Cache for Intermittent Computing Systems with Non-Volatile Main Memory

ASPLOS '25

ACM International Conference on Architectural Support for Programming Languages and Operating Systems

Mar. 30 - Apr. 3, 2025

SOURAV MOHAPATRA, **VITO KORTBEEK**, MARCO ANTONIO VAN EERDEN, JOCHEM BROEKHOFF, SAAD AHMED, PRZEMYSŁAW PAWEŁCZAK

❶ NACHO integrates write-after-read detection into a data cache and reduced runtime overhead by 54% on average.

📄 <https://doi.org/10.1145/3676641.3715989>

🔗 <https://github.com/TUDSSL/intermittent-risc-v/tree/nacho-artifact-release>

WARio: Efficient Code Generation for Intermittent Computing

PLDI '22

ACM SIGPLAN Conference on Programming Language Design and Implementation

Jun. 13 - 17, 2022

VITO KORTBEEK, SOURADIP GHOSH, JOSIAH HESTER, SIMONE CAMPANONI, PRZEMYSŁAW PAWEŁCZAK

❶ Compiler transformations that reduced checkpoint overhead by 58% on average and up to 88%.

📄 <https://doi.org/10.1145/3519939.3523454>

🔗 <https://github.com/TUDSSL/WARio>

BFree: Enabling Battery-free Sensor Prototyping with Python

UbiComp '21

ACM International Joint Conference on Pervasive and Ubiquitous Computing

Sep. 21 - 26, 2021

VITO KORTBEEK, ABU BAKAR, STEFANY CRUZ, KASIM SINAN YILDIRIM, PRZEMYSŁAW PAWEŁCZAK, JOSIAH HESTER

❶ A Python-based platform for prototyping battery-free sensors.

📄 <https://doi.org/10.1145/3432191>

🔗 <https://github.com/TUDSSL/BFree>

🌐 Adafruit, Hackaday, The Independent, de Volkskrant

Battery-Free Game Boy

UbiComp '20

ACM International Joint Conference on Pervasive and Ubiquitous Computing

Sep. 12 - 17, 2020

JASPER DE WINKEL, **VITO KORTBEEK**, JOSIAH HESTER, PRZEMYSŁAW PAWEŁCZAK

❶ An energy-harvesting handheld gaming platform that operates without batteries.

📄 <https://doi.org/10.1145/3411839> 🏆 **Distinguished Paper Award**

🔗 <https://github.com/TUDSSL/ENGAGE>

🌐 The Wall Street Journal, The Verge, CNET, Engadget

Time-sensitive Intermittent Computing Meets Legacy Software

ASPLOS '20

ACM International Conference on Architectural Support for Programming Languages and Operating Systems

Mar. 16 - 20, 2020

VITO KORTBEEK, KASIM SINAN YILDIRIM, ABU BAKAR, JACOB SORBER, JOSIAH HESTER, PRZEMYSŁAW PAWEŁCZAK

❶ Compiler and runtime support for running legacy embedded software intermittently.

📄 <https://doi.org/10.1145/3373376.3378476>

🔗 <https://github.com/TUDSSL/TICS>

Earlier Projects

Head Communications, ESA Business Incubation Centre

Noordwijk, the Netherlands

WIRELESS SENSOR SYSTEMS INTERN (BSC GRADUATION PROJECT)

Feb. 2015 - Sep. 2015

GPS, GSM, and Bluetooth LE connectivity; Embedded Linux; ZeroMQ-based data distribution; Android tooling.

HvA CleanMobility: Shell Eco-marathon and Solar Boat Challenge

Amsterdam, the Netherlands

EMBEDDED HARDWARE AND SOFTWARE DEVELOPER (EXTRACURRICULAR PROJECT)

Sep. 2012 - Aug. 2013

OLED display; power-failure-resilient data logging; GPS; brushed and brushless motor control; telemetry; first-place H2A vehicle.

Skills

Compiler development	LLVM/Clang middle end, memory-dependence analysis, transformations, retargetable compilers
Programming	C, C++, Python, Assembly, Bash, Java, Lua, VHDL
Architectures	RISC-V, ARM, MSP430, ATmega, many custom ASIP and DSP architectures
Tools	Git, CMake, Make, GDB, Docker, GNU/Linux, Embedded Linux, RTOS
Languages	Dutch (native), English (C1; IELTS 8)